COMPUTER NETWORKS · INTERVIEW NOTES

Computer Networks *Detailed* Interview Notes

The full interview-depth version — every topic a placement / SDE interview probes, with worked examples, formulas, tables, mnemonics and rapid-fire Q&A. Corrected and colour-coded.

22	95+	16	18
SECTIONS	TOPICS	FORMULA GROUPS	PRACTICE QS

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01 Network Fundamentals

DEFINITION

A **network** is a set of two or more devices (**nodes**) connected by a physical medium (**link**) so they can share data and resources. A **protocol** is the set of rules that governs how those devices communicate.

Network	Two or more devices (called **nodes**) connected by a physical medium (a **link**) so they can share data and resources.
Node	Any device on the network — computer, phone, printer, router.
Link	The medium connecting two nodes (copper cable, optical fibre, or wireless).
Protocol	A set of rules for communication. Elements: **syntax** (format), **semantics** (meaning of bits), **timing** (when & how fast).
ISP	Internet Service Provider — gives you your Internet connection and a public IP.

The Internet is a **network of networks**: many small networks (LANs) joined by routers into a global WAN. It is **decentralised** and uses **packet switching** rather than a dedicated line.

What makes a network good?

- **Performance** — measured by transmit time and response time.
- **Reliability** — measured by frequency of failure and downtime.
- **Robustness** — how well it survives faults or attacks.
- **Security** — protection of data from unauthorised access.

Communication models

Client-Server	A central **server** provides a service; many **clients** request it. Easy to manage and secure, but the server is a single point of failure and can bottleneck.
Peer-to-Peer (P2P)	Every node acts as both client and server — no central authority. Scalable and cheap, but harder to secure (e.g. BitTorrent).

02 Network Topologies

DEFINITION

Network topology is the physical or logical **layout** in which devices and links are arranged — for example star, ring, bus, mesh or hybrid.

Topology = the physical or logical **layout** of how devices and cables are connected.

TOPOLOGY	HOW IT IS WIRED	PROS	CONS
Star	All nodes connect to one central device (hub/switch).	Easy to install, manage, troubleshoot; one cable failure affects only one node.	Central device failure brings the whole network down.
Ring	Each node connects to two neighbours, forming a closed loop.	No central server needed; orderly data flow.	One node failure breaks the loop; costly, rarely used (SONET, SDH).
Bus	All nodes share a single backbone cable.	Cheap, good for few devices.	Backbone failure kills the network; collisions on shared medium.
Mesh	Every node links to many others (full or partial).	Very robust; no central dependency.	High cabling cost and complex setup.
Tree	Star networks joined to a common bus ("expanded star").	Segments maintained independently; scalable.	Depends on the main bus — if it breaks, everything fails.
Hybrid	A mix of two or more topologies.	Flexible, adapts to any environment.	Complex and expensive to design.

FORMULA — FULL MESH

Links = $n(n-1)/2$ · ports per node = $n-1$

EXAMPLE

5 nodes fully connected → 5×4/2 = **10 links**. That is why full mesh is only used for small, critical networks.

REMEMBER

Star = central device is the single point of failure. **Mesh** = most reliable, most expensive. Most modern LANs use **star** wiring.

03 Types of Networks

DEFINITION

Networks are classified by the **geographical area** they cover — from a personal area (PAN) up to a global area network (GAN).

Classified by **geographical area** covered.

TYPE	FULL FORM	RANGE / USE
PAN	Personal Area Network	Up to ~10 m — personal devices (phone, earbuds, printer).
LAN	Local Area Network	A small area — office, home, school. High speed, privately owned.
HAN	House Area Network	A LAN within one house.
CAN	Campus Area Network	Links departments within one campus.
MAN	Metropolitan Area Network	A city — spans a metro area.
WAN	Wide Area Network	Very large — connects cities and countries. The Internet is the biggest WAN.
GAN	Global Area Network	Global coverage, often via satellites.

VPN (Virtual Private Network)

A private network built **on top of the public Internet** using an encrypted tunnel. A remote client can safely reach an organisation's network, and it is far cheaper than dedicated WAN lines.

- **Access VPN** — for remote/mobile users (replaces dial-up).
- **Site-to-Site VPN** — links two office networks (Intranet & Extranet variants).

DON'T CONFUSE THESE

Internet = public, for everyone. **Intranet** = private, inside one organisation. **Extranet** = a controlled part of an intranet shared with partners.

04 Network Devices

DEFINITION

Network devices are the hardware components (hub, switch, router, gateway, modem, NIC) that connect nodes and move data between them.

DEVICE	LAYER	WHAT IT DOES
Repeater	Physical (L1)	Regenerates a weak signal so it can travel farther.
Hub	Physical (L1)	Blindly repeats a signal to every port except the incoming one. No filtering — causes collisions.
Bridge	Data Link (L2)	Joins two LAN segments and filters traffic by MAC address (a 2-port switch).
Switch	Data Link (L2)	Forwards frames only to the correct port using MAC addresses. Full-duplex, "intelligent bridge".
Router	Network (L3)	Connects different networks and chooses the best path using IP addresses.
Gateway	Any	A node connecting two *dissimilar* networks (a router connects *similar* ones).
Modem	L1/L2	Converts digital data to/from the signals used on the ISP line.
NIC	L1/L2	Network Interface Card — hardware that joins a PC to a network; carries its unique MAC address.
Access Point	L2	Bridges wireless devices onto a wired LAN (Wi-Fi).
Firewall	L3–L7	Monitors traffic and blocks it per security policy. Types: **packet-filtering** (stateless), **stateful**, **proxy**, and **NGFW** (deep inspection).

ONE-LINER TO MEMORISE

Hub floods, **switch** filters by MAC, **router** routes by IP. Layer 1 → 2 → 3.

Collision domain vs broadcast domain

- A **collision domain** is where two frames can collide. Each switch port is its own collision domain; a hub makes all its ports one domain.
- A **broadcast domain** is the set of devices that receive a broadcast. Switches keep one broadcast domain (unless split by VLANs); **routers** separate broadcast domains.

05 Switching Techniques

DEFINITION

Switching is the technique of forwarding data from a source to a destination across a network. The two main types are **circuit switching** (a dedicated path) and **packet switching** (independent packets).

How data is moved from source to destination across a network.

	CIRCUIT SWITCHING	PACKET SWITCHING
Path	A dedicated path is set up *before* communication.	No dedicated path; each packet is routed independently.
Bandwidth	Reserved and guaranteed	Shared, best-effort
Delay	Fixed after setup; setup itself takes time	Variable (depends on traffic)
Efficiency	Wasteful when the line is idle	Efficient — links stay busy
Example	Traditional telephone network	The Internet

Packet switching: two flavours

- **Datagram** — every packet carries the full destination address and can take any path (used by IP). No order guarantee.
- **Virtual circuit** — a path is fixed first, then packets follow it with a short label (e.g. MPLS, Frame Relay).

Message switching — the whole message is stored and forwarded at each node. Needs huge storage, so it is rarely used today.

06 Transmission Modes & Media

DEFINITION

The **transmission mode** describes the direction of data flow (simplex, half-duplex, full-duplex); the **transmission medium** is the physical channel that carries the signal (guided or unguided).

Transmission modes — direction of data flow

- **Simplex** — one direction only (keyboard → computer, TV broadcast).
- **Half-duplex** — both directions, but only one at a time (walkie-talkie).
- **Full-duplex** — both directions simultaneously (telephone, modern switch).

Multiplexing — many signals over one link

- **FDM** — Frequency Division (each signal gets a frequency band).
- **TDM** — Time Division (each signal gets a time slot).
- **WDM** — Wavelength Division (used in optical fibre).

Transmission media — the physical channel

MEDIA	TYPE	NOTES
Twisted pair (UTP/STP)	Guided	Cheapest, used in Ethernet (Cat5e/Cat6). Susceptible to interference.
Coaxial cable	Guided	Used in cable TV / older Internet. Better shielding than UTP.
Optical fibre	Guided	Fastest; immune to electromagnetic interference. Single-mode (long distance) & multi-mode (short).
Radio / Microwave / Infrared	Unguided	Wireless — no cable. Used by Wi-Fi, cellular, satellite.

07 The OSI Model

DEFINITION

The **OSI (Open Systems Interconnection) model** is a 7-layer conceptual framework, defined by ISO, that standardises how two systems communicate over a network.

OSI (Open Systems Interconnection) is a 7-layer reference model from **ISO** (1984) that standardises how any two systems communicate. Each layer has one well-defined job and talks only to the layers directly above and below it.

```
Mnemonic (top→bottom): All People Seem To Need Data Processing  ·
(bottom→top): Please Do Not Throw Sausage Pizza Away
```

#	LAYER	JOB	PDU	PROTOCOLS / DEVICES
7	**Application**	User-facing services	Data	HTTP, FTP, SMTP, DNS
6	**Presentation**	Translation, encryption, compression	Data	SSL/TLS, JPEG, ASCII
5	**Session**	Start / maintain / end sessions	Data	RPC, NetBIOS
4	**Transport**	End-to-end delivery, flow & error control	Segment	TCP, UDP · adds *ports*
3	**Network**	Logical addressing & routing	Packet	IP, ICMP, ARP · **router**
2	**Data Link**	Framing, MAC addressing, error control	Frame	Ethernet, MAC · **switch**
1	**Physical**	Raw bit transmission over the medium	Bits	Cables, radio · **hub**

INTERVIEW ANGLE

Be ready to name, for any attack, which layer it hits — e.g. ARP spoofing = L2, IP spoofing = L3, SYN flood = L4, SQL injection = L7.

Q. Why do we need a layered model at all?

It splits a complex problem into independent, replaceable pieces — e.g. you can swap Wi-Fi for Ethernet (L1/L2) without touching HTTP (L7).

08 The TCP/IP Model

> **DEFINITION**
>
> The **TCP/IP model** is a 4-layer practical framework describing how data actually travels across the Internet; it is the model the Internet is built on.

The practical model the Internet actually runs on — a compressed 4-layer version of OSI, developed by the **US Department of Defense (DARPA)** in the **1970s**.

TCP/IP LAYER	JOB	PROTOCOLS	MAPS TO OSI
Application	User services & data	HTTP, DNS, SMTP, FTP	App + Presentation + Session
Transport	End-to-end, ports	TCP, UDP	Transport
Internet	Addressing & routing	IP, ICMP, ARP	Network
Link / Network Access	Physical delivery	Ethernet, SONET	Data Link + Physical

OSI vs TCP/IP — the standard answer

- OSI has **7** layers (a theoretical reference); TCP/IP has **4** (practical, actually used).
- OSI was protocol-independent; TCP/IP was built *around* its protocols (TCP and IP).
- OSI separates Session/Presentation; TCP/IP folds them into Application.

> **COMMON MISTAKE**
>
> Some notes say TCP/IP was created in the **1860s** — wrong. It is the **1970s** (ARPANET / DoD work).

09 Encapsulation & Data Flow

DEFINITION

Encapsulation is the process of adding a header (and sometimes a trailer) to data as it moves *down* the layers, so each layer can deliver it correctly; the reverse is **de-encapsulation**.

As data moves **down** the sender's layers, each layer adds its own header — this is **encapsulation**. At the receiver the headers are stripped off on the way **up** — **de-encapsulation**.

Data	Application layer produces the message.
Segment	Transport adds a TCP/UDP header (ports, sequence numbers).
Packet	Network adds an IP header (source & destination IP).
Frame	Data Link adds a MAC header + trailer (FCS for error check).
Bits	Physical converts it all into signals on the wire.

SAY IT LIKE THIS

Data → Segment → Packet → Frame → Bits. That single line answers a surprising number of questions.

Q. What is MTU and why does fragmentation happen?

MTU (Maximum Transmission Unit) is the largest frame a link can carry (Ethernet ≈ 1500 bytes). If a packet is bigger, the network layer splits it into fragments — that is **fragmentation**.

10 IP Addressing

> **DEFINITION**
> An **IP address** is a unique logical address assigned to a device on a network, used to identify it and route data to it. IPv4 is 32-bit; IPv6 is 128-bit.

An **IPv4 address** is a **32-bit** address written as four 8-bit octets, each 0–255 (e.g. `192.168.1.1`). It has a **network ID** and a **host ID** part.

CLASS	FIRST OCTET	DEFAULT MASK	USED FOR
A	1–126	/8	Very large networks (16M hosts)
B	128–191	/16	Medium networks (65K hosts)
C	192–223	/24	Small networks (254 hosts)
D	224–239	—	Multicast
E	240–255	—	Experimental

FORMULA

Hosts in a class = $2^{(\text{host bits})} - 2$ · e.g. Class C → $2^8-2 = 254$

Private vs Public IP

- **Private IPs** are reserved and *not* routable on the Internet: `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`. To reach the Internet they go through **NAT**.
- **Public IPs** are assigned by the ISP and are globally unique.
- **Loopback** = `127.0.0.1` (your own machine). **APIPA** = `169.254.x.x` (self-assigned when DHCP fails).

Static vs Dynamic IP — static is fixed (servers, printers); dynamic is handed out on demand by **DHCP**.

NAT (Network Address Translation)

Rewrites a private IP to a public IP so many internal devices can share one public address.

- **Static NAT** — one private ↔ one public (1-to-1).
- **Dynamic NAT** — maps to a pool of public IPs.
- **PAT / NAT-overload** — many-to-one using different ports (what your home router does).

IPv4 vs IPv6

	IPV4	IPV6
Size	32-bit (~4.3 billion addresses)	128-bit (practically unlimited)
Format	Dotted decimal	Hexadecimal, colon-separated
Security	Optional (IPsec)	Built-in (IPsec mandatory)
Header	Variable, with checksum	Fixed 40 bytes, simpler

11 Subnetting & CIDR

DEFINITION

Subnetting is the process of dividing a network into smaller sub-networks; **CIDR** is the `/n` notation that specifies how many leading bits form the network prefix.

Subnetting splits one network into smaller sub-networks. It improves routing efficiency, reduces broadcast traffic, and improves security.

Subnet mask — A 32-bit number that separates the network part from the host part (e.g. `255.255.255.0`).

CIDR — Classless Inter-Domain Routing — the `/n` notation. `/24` = 24 network bits, 8 host bits.

VLSM — Variable Length Subnet Masking — using different mask sizes inside one network to avoid wasting addresses.

FORMULAS

Hosts per subnet = $2^{(32-prefix)} - 2$

Number of subnets = 2^n (n = bits borrowed)

New prefix = old prefix + n

PREFIX	SUBNET MASK	ADDRESSES	USABLE HOSTS
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/30	255.255.255.252	4	2

WORKED EXAMPLE 1

Split `192.168.1.0/24` into **4 subnets**.

- 4 subnets need 2 bits ($2^2 = 4$) → new prefix `/26`.
- Each subnet = $2^{(32-26)}$ = **64 addresses** → **62 usable**.
- Subnets: `.0–.63`, `.64–.127`, `.128–.191`, `.192–.255`.
- Subnet 1: network `192.168.1.0`, first host `.1`, last host `.62`, broadcast `.63`.

WORKED EXAMPLE 2

How many hosts fit in `10.0.0.0/8`? → host bits = 32−8 = 24 → $2^{24}-2$ = **16,777,214 hosts**.

12 MAC vs IP Address

DEFINITION

A **MAC address** is a 48-bit physical address burnt into a NIC by its manufacturer; an **IP address** is a logical, routable address assigned by software (DHCP/ISP).

	MAC ADDRESS	IP ADDRESS
Type	Physical address	Logical address
Given by	NIC manufacturer (burnt in)	ISP / DHCP
Layer	Data Link (L2)	Network (L3)
Format	48-bit hex (e.g. `00:1A:2B:3C:4D:5E`)	32-bit IPv4 / 128-bit IPv6
Scope	Local network only	Global (routable)
Changes?	Fixed (unless spoofed)	Can change (DHCP)

A MAC address is 48 bits: the first 24 bits are the **OUI** (manufacturer ID), the last 24 bits are the device's serial number.

ARP — how IP finds MAC

ARP (Address Resolution Protocol) maps a known IP address to the corresponding MAC address on a local network.

1. The sender broadcasts: "Who has `192.168.1.5`?"
2. The owner replies with its MAC address.
3. The sender caches the mapping in its **ARP cache** for a short time.

RARP does the reverse (MAC → IP).

13 Transport Layer: TCP & UDP

DEFINITION

The **transport layer** provides end-to-end communication between processes on two hosts using **ports**. **TCP** is reliable and connection-oriented; **UDP** is fast and connectionless.

	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable — acknowledgements & retransmission	Unreliable — best effort
Speed	Slower (overhead)	Faster, lightweight
Ordering	Guaranteed in-order	No ordering guarantee
Header size	20–60 bytes (large)	8 bytes (small)
Flow / congestion control	Yes	No
Used by	HTTP/HTTPS, FTP, SMTP, SSH	DNS, DHCP, streaming, gaming, VoIP

The TCP 3-Way Handshake

1. **SYN** — client asks to connect (with its sequence number).
2. **SYN-ACK** — server agrees and acknowledges.
3. **ACK** — client confirms; connection established.

Closing takes a similar **4-way** exchange (FIN → ACK → FIN → ACK).

TCP header fields

Source port, destination port, **sequence number**, **acknowledgement number**, flags, window size, checksum, urgent pointer. The **window size** enables flow control.

TCP flags: **SYN** (start), **ACK** (acknowledge), **FIN** (finish), **RST** (reset/abort), **PSH** (push), **URG** (urgent).

SECURITY LINK

A **SYN flood** abuses the handshake — the attacker sends many SYNs but never the final ACK, exhausting the server's half-open connections.

14 Error Detection & Flow Control

DEFINITION

Error detection finds corrupted bits during transmission (parity, checksum, CRC); **flow control** prevents a fast sender from overwhelming a slow receiver.

Error detection — catching corrupted bits

Parity bit	Adds one extra bit to make the count of 1s odd or even. Detects single-bit errors only.
Checksum	Sums the data and sends the total; the receiver re-checks. Used by IP, TCP and UDP.
CRC	Cyclic Redundancy Check — polynomial division. The strongest of the three; used by Ethernet.

Flow control — protecting a slow receiver

- **Stop-and-Wait** — send one frame, wait for its ACK, then send the next. Simple but slow.
- **Sliding Window** — send several frames before waiting. Variants: *Go-Back-N* (resend from the lost frame) and *Selective Repeat* (resend only the lost frame).

Congestion control (TCP)

Manages *network* overload: **slow start** (grow the window exponentially), **congestion avoidance**, and **AIMD** (additive increase, multiplicative decrease).

Media access methods

- **CSMA/CD** — Collision *Detection*; listen before sending, detect collisions, back off. Wired Ethernet.
- **CSMA/CA** — Collision *Avoidance*; used in Wi-Fi where collisions can't be detected reliably.

Q. Difference between flow control and congestion control?

Flow control stops a fast *sender* overwhelming a slow *receiver*; congestion control stops senders overwhelming the *network* itself.

15 Application-Layer Protocols

DEFINITION

Application-layer protocols define how specific user services exchange data — for example HTTP (web), DNS (names), SMTP (email) and FTP (files).

DNS	Domain Name System — translates names (`google.com`) into IP addresses. Paul Mockapetris & Jon Postel (1983). Runs on **UDP port 53**.
DHCP	Dynamic Host Configuration Protocol — auto-assigns IP, subnet mask and DNS. **Port 67** (server) / 68 (client).
HTTP / HTTPS	Rules for transferring web pages. HTTPS = HTTP over TLS/SSL. Ports **80** / **443**.
FTP	File Transfer Protocol — control on **port 21**, data on **port 20**. SFTP is the secure version (over SSH, port 22).
SMTP	Simple Mail Transfer Protocol — *sends* email. **Port 25**.
POP3 / IMAP	Retrieve email. POP3 (**110**) downloads & often deletes; IMAP (**143**) syncs and keeps mail on the server.
ICMP	Internet Control Message Protocol — error reporting & diagnostics (`ping` , `traceroute`). Network-layer, so it has **no ports**.
ARP	Address Resolution Protocol — maps IP → MAC on a local network.
RIP / OSPF / BGP	Routing protocols. RIP = hop count; OSPF = link-state (inside an organisation); BGP = the Internet backbone between ISPs.
SSH / Telnet	Remote login. SSH (**22**) is encrypted; Telnet (**23**) is plain-text and insecure.

DNS record types COMMONLY ASKED

RECORD	MEANING
A	Maps a name to an IPv4 address
AAAA	Maps a name to an IPv6 address
CNAME	An alias pointing to another name
MX	Mail server for the domain
NS	Authoritative name server
TXT	Free text — verification & SPF
PTR	Reverse lookup (IP → name)

DNS resolution — step by step

Browser cache → OS cache → the *recursive resolver* (often the ISP) → **root** server → **TLD** server (`.com`) → **authoritative** server → the IP is returned and cached for a **TTL**.

Recursive query = the resolver fetches the full answer; **iterative** query = each server just points you to the next.

DHCP DORA process

How a device gets its IP: **D**iscover → **O**ffer → **R**equest → **A**cknowledge.

Email flow

Sender's client sends via **SMTP** to the sender's server → SMTP delivers to the recipient's server → the recipient fetches it with **POP3/IMAP**.

16 Port Numbers

DEFINITION

A **port number** is a 16-bit number that identifies a specific process or service on a host, allowing one IP address to run many services at once.

Ports identify a specific **process** on a host. Range: **0–1023** well-known, **1024–49151** registered, **49152–65535** dynamic/private.

PORT	PROTOCOL	PURPOSE
20 / 21	FTP	File transfer (data / control)
22	SSH / SFTP	Secure remote login / file transfer
23	Telnet	Remote login (insecure)
25	SMTP	Sending email
53	DNS	Name resolution
67 / 68	DHCP	Automatic IP assignment
80	HTTP	Web (unencrypted)
110	POP3	Receiving email
143	IMAP	Receiving email (synced)
161	SNMP	Network device management
443	HTTPS	Web (encrypted)
3306	MySQL	Database
3389	RDP	Remote Desktop
8080	HTTP-alt	Alternate web / proxy

WHY PORTS MATTER

One IP can run many services at once — the port tells the OS which process a packet belongs to. A connection is uniquely identified by the **4-tuple**: source IP + source port + destination IP + destination port.

17 HTTP & the Web

> DEFINITION
>
> **HTTP (HyperText Transfer Protocol)** is a **stateless** application-layer protocol that defines how clients and servers exchange web requests and responses.

HTTP is a **stateless** application-layer protocol — each request is independent of the previous one. It runs over TCP.

Methods

GET (read), **POST** (create), **PUT** (replace), **PATCH** (partial update), **DELETE** (remove), **HEAD**, **OPTIONS**. GET is *idempotent* & cacheable; POST is not.

Status codes

CODE	MEANING
`200`	OK — success
`201`	Created
`301 / 302`	Moved permanently / temporarily (redirect)
`304`	Not Modified (use cache)
`400`	Bad Request
`401 / 403`	Unauthorized / Forbidden
`404`	Not Found
`429`	Too Many Requests (rate-limited)
`500 / 502 / 503`	Server error / Bad gateway / Unavailable

Classes: **1xx** info, **2xx** success, **3xx** redirect, **4xx** client error, **5xx** server error.

Cookies vs Sessions vs Tokens

- **Cookies** — small data stored in the *browser*, sent with every request.
- **Sessions** — state stored on the *server*; the cookie just holds a session ID.
- **Tokens (JWT)** — a signed token the client stores and sends; the server verifies it without storing session state (stateless auth).

Common headers: `Host`, `User-Agent`, `Content-Type`, `Accept`, `Cookie`, `Authorization`, `Cache-Control`.

HTTPS & TLS handshake (simplified)

1. Client hello → server hello, and the server sends its **certificate**.
2. The client verifies the certificate (via a Certificate Authority) and both exchange keys.
3. Both derive a shared **session key**; from now on traffic is encrypted.

HTTP versions

HTTP/1.1 — one request at a time per connection. **HTTP/2** — multiplexing over one connection. **HTTP/3** — runs over QUIC (UDP) for lower latency.

REST — an architectural style for APIs using HTTP methods, resource URLs and stateless requests.

18 Performance & Content Delivery

DEFINITION

Latency is the delay for data to travel; **bandwidth** is the link's capacity; **throughput** is the actual data rate achieved (always ≤ bandwidth).

Latency	Time for a packet to travel from source to destination (delay).
Bandwidth	Maximum data-carrying capacity of a link (the "width of the pipe").
Throughput	The *actual* data transferred per second — always ≤ bandwidth.
Jitter	Variation in latency — bad for voice/video.
Packet loss	Packets that never arrive; forces retransmission.

CLASSIC ANALOGY

Bandwidth is the *number of lanes* on a highway; latency is *how long one car takes*. More lanes (bandwidth) don't shorten the road (latency).

Scaling & delivery

- **Load Balancer** — spreads traffic across servers. **L4** balances by IP/port; **L7** by URL/header content.
- **CDN** — edge servers cache content near users, cutting latency.
- **Forward proxy** — in front of *clients* (hides the client). **Reverse proxy** — in front of *servers* (hides the server, adds caching/SSL/load-balancing).

19 Classic Interview Questions

"What happens when you type `google.com` and press Enter?" `MOST ASKED`

1. The browser checks its **cache**; if the page is fresh, it renders it and stops.
2. Otherwise it checks the browser/OS cache for the IP; if missing, the OS does a **DNS lookup** (using UDP) to translate the domain into an IP.
3. A **TCP 3-way handshake** (SYN / SYN-ACK / ACK) opens a connection to the server.
4. For HTTPS, a **TLS handshake** secures the channel.
5. An **HTTP request** is sent; the server returns an **HTTP response**.
6. The browser renders the response (or caches it), and the TCP connection is closed or reused.

Rapid-fire Q&A

Q. What is a socket?

An endpoint of communication — the combination of an **IP address + a port number** (e.g. `192.168.1.5:443`). A connection is a pair of sockets.

Q. Hub vs Switch vs Router?

Hub = L1, floods to all ports. Switch = L2, forwards by MAC. Router = L3, routes between networks by IP.

Q. Difference between latency and bandwidth?

Latency is the delay of a single packet; bandwidth is the capacity of the link. More bandwidth does not reduce latency.

Q. What is NAT and why is it used?

It translates private IPs to a public IP so many devices share one public address — conserving IPv4 addresses.

Q. Forward proxy vs reverse proxy?

A forward proxy fronts clients (hides them); a reverse proxy fronts servers (hides them, adds caching/SSL/load-balancing).

Q. What is APIPA?

A self-assigned `169.254.x.x` address a host gives itself when DHCP fails.

Addressing modes

- **Unicast** — one sender → one receiver.
- **Broadcast** — one sender → all nodes (ARP & DHCP use this).
- **Multicast** — one sender → a group of nodes.
- **Anycast** — one sender → the nearest of several nodes (CDNs).

Handy commands: `ping` (connectivity), `traceroute` (path), `ipconfig` (Windows) / `ifconfig` (Linux/Mac), `netstat` (connections).

20 Numericals & Formulas

1 · Subnetting

FORMULAS

Hosts per subnet = $2^{(32-\text{prefix})} - 2$

Number of subnets = 2^{n} (n = borrowed bits)

New prefix = old prefix + n

EXAMPLE

Need **8 subnets** from `192.168.10.0/24`. $2^3 = 8$ → borrow 3 bits → new prefix `/27`, mask `255.255.255.224`. Hosts per subnet = $2^5 - 2 = \mathbf{30}$. Ranges: `.0–.31`, `.32–.63`, `.64–.95`, …

2 · Hosts per IP class

FORMULA

Hosts = $2^{(\text{host bits})} - 2$

EXAMPLES

- Class A: $2^{24} - 2 = \mathbf{16{,}777{,}214}$
- Class B: $2^{16} - 2 = \mathbf{65{,}534}$
- Class C: $2^{8} - 2 = \mathbf{254}$

3 · Transmission delay (T_t)

FORMULA

T_t = **Message size (bits) ÷ Bandwidth (bps)**

EXAMPLE

Send a **1 MB** file over a **1 Mbps** link: 1 MB = 8×10^6 bits. T_t = 8,000,000 ÷ 1,000,000 = **8 seconds**.

4 · Propagation delay (T_p)

FORMULA

T_p = **Distance ÷ Propagation speed**

EXAMPLE

A signal travels **3000 km** over fibre at 2×10^8 m/s. T_p = 3,000,000 ÷ 200,000,000 = **0.015 s = 15 ms**.

5 · Total delay & RTT

FORMULAS

One-way delay = $T_t + T_p$ (+ queuing & processing)

RTT ≈ $2 \times T_p$

EXAMPLE

Using the values above (T_t=8 s, T_p=15 ms): one-way delay ≈ 8.015 s; RTT ≈ 2×15 = **30 ms**.

6 · Bandwidth-Delay Product (BDP)

FORMULA

BDP = **Bandwidth × RTT** (bits "in flight")

EXAMPLE

1 Mbps × 30 ms = 1,000,000 × 0.03 = **30,000 bits** ≈ 3.75 KB in the pipe at any moment.

7 · Throughput

FORMULA

$$\text{Throughput} = \text{Data transferred} \div \text{Time}$$

EXAMPLE

5 MB transferred in 4 s = 1.25 MB/s = **10 Mbps**.

8 · Stop-and-Wait efficiency

FORMULA

$$\text{Efficiency} = T_t \div (T_t + 2T_p)$$

EXAMPLE

$T_t = 8$ s, $T_p = 15$ ms → $8 \div (8 + 0.03) \approx$ **99.6%**. Short frames over long links have poor efficiency — that is why sliding window is used.

9 · Channel capacity (advanced)

FORMULAS

$$\text{Nyquist: } C = 2B \cdot \log_2(M) \quad \text{(noiseless)}$$

$$\text{Shannon: } C = B \cdot \log_2(1 + SNR) \quad \text{(noisy)}$$

EXAMPLE

B = 3 kHz, M = 8 levels → $C = 2 \times 3000 \times \log_2 8 =$ **18,000 bps**.

10 · Sliding window (sequence numbers & efficiency)

FORMULAS

Sequence numbers — Stop-and-Wait = **2**, Go-Back-N = **N+1**, Selective Repeat = **2N**

Max window for 100% use: $W = 1 + 2a$, where $a = T_p \div T_t$

Efficiency = $W \times T_t \div (T_t + 2T_p)$ (when $W \le 1+2a$)

EXAMPLE

T_t = 1 ms, T_p = 20 ms → a = 20 → W = 1 + 40 = **41 frames** for full utilisation. With W = 10, efficiency = 10×1 ÷ (1 + 40) ≈ **24%**.

11 · Fragmentation

FORMULAS

Data per fragment = **MTU − IP header** (usually 1500 − 20 = 1480 bytes)

No. of fragments = **ceil(payload ÷ data-per-fragment)**

EXAMPLE

A 3000-byte payload over a 1500-byte MTU → 1480 + 1480 + 40 = **3 fragments**. Fragment offsets (in 8-byte units) = 0, 185, 370; the MF flag is set on all but the last.

12 · CRC (error detection)

METHOD

Append **r zeros** to the data (r = degree of the generator), divide by the generator using **XOR**; the remainder is the CRC. Send data + remainder.

EXAMPLE

Data `1101`, generator `1011` (r = 3). Append 000 → 1101000 ÷ 1011 (XOR division) → remainder **001**. Codeword sent = `1101001`.

13 · Checksum

METHOD

Add all the data words (1's complement addition), wrap the carry around, then take the **1's complement** of the sum.

EXAMPLE

Words `10110011` + `01100101` = 1 00011000 → wrap the carry → 00011001 → 1's complement = **11100110** (the checksum).

14 · Hamming code (error correction)

FORMULA

Redundant bits: $2^r \geq m + r + 1$ (m = data bits)

EXAMPLES

m = 4 → $2^3 = 8 \geq 8$ → r = **3** parity bits (7-bit code). m = 8 → $2^4 = 16 \geq 13$ → r = **4** (12-bit code).

15 · Supernetting (route aggregation)

METHOD

Combine contiguous subnets by **reducing the prefix length** (moving the mask left) so they share one network address.

EXAMPLE

192.168.0.0/24, .1.0/24, .2.0/24, .3.0/24 → aggregate to **192.168.0.0/22** (covers .0 – .3).

16 · Time to send N packets (pipelining)

FORMULA

Total time = $N \times T_t + T_p$ (pipelined, ignoring ACKs)

EXAMPLE

100 packets, T_t = 1 ms, T_p = 10 ms → 100×1 + 10 = **110 ms** (vs 2100 ms with stop-and-wait).

21 Practice Numericals (Solved)

Q1 **A company has the network `192.168.10.0/24` and wants subnets that each support at least 50 hosts. Find the subnet mask, the number of subnets, and the range of the first subnet.**

SOLUTION

- Need $2^h - 2 \geq 50$ → h = 6 gives 62 hosts ✓ (h = 5 gives only 30 ✗)
- Host bits = 6 → prefix = 32 − 6 = **/26** → mask = `255.255.255.192`
- Number of subnets = $2^{(26-24)}$ = **4**
- First subnet `192.168.10.0/26` : hosts `.1` – `.62` , broadcast `.63`
- **Mask /26, 4 subnets, first range .1 – .62**

Q2 **From `10.0.0.0/8` you need at least 1000 subnets. How many bits must you borrow, what is the new prefix, and how many hosts per subnet?**

SOLUTION

- $2^n \geq 1000$ → n = 10 ($2^{10} = 1024$) ✓
- New prefix = 8 + 10 = **/18**
- Host bits = 32 − 18 = 14 → hosts = $2^{14} - 2$ = **16,382**
- **Borrow 10 bits, prefix /18, 16,382 hosts per subnet**

Q3 **How many usable host addresses does a Class C network such as `200.1.1.0` have?**

SOLUTION

- Class C → 24 network bits, 8 host bits
- Hosts = $2^8 - 2 = 256 - 2$
- **= 254 usable hosts**

Q4 **A 5 MB file is sent over a 10 Mbps link. Find the transmission delay.**

SOLUTION

- Size = 5 MB = $5 \times 8 \times 10^6$ = 40,000,000 bits
- T_t = size ÷ bandwidth
- T_t = 40,000,000 ÷ 10,000,000
- **= 4 seconds**

Q5 **A signal travels 5000 km over fibre at 2×10^8 m/s. Find the propagation delay.**

SOLUTION

- Distance = 5000 km = 5.000.000 m
- T_p = distance ÷ speed
- T_p = 5,000,000 ÷ 200,000,000 = 0.025 s
- **= 25 ms**

Q6 **On a link, T_t = 1 ms and T_p = 8 ms. Find the one-way delay and the RTT.**

SOLUTION

- One-way delay = $T_t + T_p = 1 + 8 =$ **9 ms**
- RTT ≈ $2 \times T_p = 2 \times 8$
- **One-way = 9 ms, RTT = 16 ms**

Q7 **A link has 100 Mbps bandwidth and an RTT of 40 ms. Find the bandwidth-delay product.**

SOLUTION

- BDP = bandwidth × RTT
- = 100,000,000 × 0.04 = 4,000,000 bits
- **= 4 Mbit = 500 KB in the pipe at any instant**

Q8 **A 20 MB file downloads in 16 seconds. Find the throughput in Mbps.**

SOLUTION

- Data = 20 MB = 20 × 8 = 160 Mbit
- Throughput = 160 ÷ 16
- **= 10 Mbps (i.e. 1.25 MB/s)**

Q9 **Stop-and-Wait: T_t = 2 ms, T_p = 18 ms. Find the efficiency.**

SOLUTION

- $\text{Efficiency} = T_t \div (T_t + 2T_p)$
- $= 2 \div (2 + 36) = 2 \div 38$
- **≈ 5.3% (very poor — why sliding window exists)**

Q10 **Sliding window: T_t = 1 ms, T_p = 15 ms. What window size gives 100% utilisation?**

SOLUTION

- $a = T_p \div T_t = 15 \div 1 = 15$
- $W = 1 + 2a$
- $W = 1 + 2 \times 15 = 31$
- **A window of 31 frames (or more)**

Q11 **Same link (T_t = 1 ms, T_p = 15 ms), but window W = 10. Find the efficiency.**

SOLUTION

- $\text{Efficiency} = W \times T_t \div (T_t + 2T_p)$
- $= 10 \times 1 \div (1 + 30) = 10 \div 31$
- **≈ 32.3%**

Q12 **A 4000-byte datagram (including a 20-byte header) is sent over a link with MTU 1500 bytes. How many fragments are formed?**

SOLUTION

- Payload = 4000 − 20 = 3980 bytes
- Data per fragment = 1500 − 20 = 1480 bytes
- 3980 ÷ 1480 = 2.69 → round up = **3 fragments**
- Payloads: 1480 + 1480 + 1020; offsets (8-byte units) = 0, 185, 370
- **3 fragments**

Q13 **Find the CRC for data `110101` using the generator `1001`.**

SOLUTION

- Degree r = 3 → append three 0s → `110101000`
- XOR-divide by `1001`; the remainder is the CRC
- Remainder = `011`
- **Codeword sent = 110101011**

Q14 **Two 8-bit words `11100010` and `00100110` are to be sent. Find the checksum.**

SOLUTION

- Sum = 11100010 + 00100110 = 1 00001000
- Wrap the carry around: 00001000 + 1 = 00001001
- 1's complement of the sum = 11110110
- **Checksum = 11110110**

Q15 **For 8 data bits, how many redundant (parity) bits does a Hamming code need?**

SOLUTION

- $2^r \geq m + r + 1$ → $2^r \geq 9 + r$
- r = 3: 8 ≥ 12 ✗ | r = 4: 16 ≥ 13 ✓
- **r = 4 parity bits (total 12-bit code)**

B1 **Bonus · Aggregate `200.10.4.0/24` through `200.10.7.0/24` into one route.**

SOLUTION

- Four contiguous /24 blocks → reduce the prefix by 2 bits ($2^2 = 4$)
- 4 → /22
- **200.10.4.0/22 (covers .4.0 – .7.255)**

B2 **Bonus · Nyquist: bandwidth 4 kHz, 16 signal levels. Find the maximum data rate.**

SOLUTION

- $C = 2B \cdot \log_2(M)$
- $= 2 \times 4000 \times \log_2(16) = 8000 \times 4$
- **= 32,000 bps = 32 kbps**

B3 **Bonus · 200 packets are sent pipelined; T_t = 2 ms, T_p = 20 ms. Find the total time.**

SOLUTION

- $Total = N \times T_t + T_p$
- = 200 × 2 + 20 = 400 + 20
- **= 420 ms (vs 8,400 ms with stop-and-wait)**

22 Quick Revision Cheat-Sheet

- OSI layers (top→bottom): **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing.
- Encapsulation: Data → Segment → Packet → Frame → Bits.
- Devices by layer: hub/repeater = L1, switch/bridge = L2, router = L3.
- TCP = reliable, ordered, slower; UDP = fast, best-effort.
- TCP handshake: SYN → SYN-ACK → ACK.
- Private ranges: `10.x`, `172.16–31.x`, `192.168.x`.
- Key ports: 22 SSH, 25 SMTP, 53 DNS, 67 DHCP, 80 HTTP, 443 HTTPS, 3306 MySQL.
- CIDR: /24 = 254 hosts, /25 = 126, /26 = 62, /27 = 30, /28 = 14.
- HTTP is stateless; state is kept with cookies / sessions / tokens.
- DNS resolves names → IP; ARP resolves IP → MAC.
- NAT lets many private devices share one public IP.
- Modes: simplex, half-duplex, full-duplex. Switching: circuit vs packet.
- Error detection: parity, checksum, CRC. Flow control: stop-and-wait, sliding window.
- DHCP = DORA. Delay: T_t = size/bw, T_p = distance/speed.

> **DO NOT REPEAT THESE WRONG ANSWERS**
>
> FTP = port **21/20** (not 27). ICMP has **no ports** (not 7). TCP/IP was designed in the **1970s** (not 1860s).